

Global scenario

The number of robots in the world today is approaching 1,000,000, with almost half that number in Japan and just 15 per cent in the US. A couple of decades ago, 90 per cent of robots were used in car manufacturing, typically on assembly lines doing a variety of repetitive tasks. Today, only 50 per cent are in automobile plants, with the other half spread out among other factories, laboratories, warehouses, energy plants, hospitals and many other industries. In the area of 'electronic business,' industrial robots are changing too. High competition existing in the global market requires 'just in time' products, better utilisation of raw materials, better utilisation of resources and lower consumption of energy. All these requirements imply new features for the industrial robots.

The robotic kits market is primarily centred around recreation, service, and learning and research.

Recreational robotics is becoming popular with robot competitions like robot soccer. This is particularly popular in Asian countries like Japan, Korea, Singapore, Taiwan and China. The current market share in this category is comparatively low now and is likely to diminish in a decade as robots become more common in our daily life and we lose excitement in their novelty. Kits in particular will find it difficult to catch this market due to their limited scope compared to research prototypes in this area.

The future trend is definitely towards service robots, which are semi or fully autonomous and can provide useful services in support of humans and manufacturing environments. There will be a substantial increase in activity and build-up of this market. Here again finished products are expected to dominate kits.

Coming to research and learning, there is a consistently growing market for robots and robotic kits in this space. As far as product development in this area is concerned, there is a need for adequate kits and spares to carry out the research and prototyping. Apart from that, robotic kits serve as a multidisciplinary learning tool for fundamentals such as mathematics, physics, and science, which again is a promising market. Here kits undoubtedly dominate any other specifically tailored product.

robots around specific applications is a crucial area. Lack of the trained system engineering manpower and reluctance of the Indian industry to open to advanced methods of manufacturing have led to a stunted demand for industrial robots.

Some of the advanced applications developed by Indian students can be seen in national-level competitions like *Micromouse* and *Robocon*. Maze solving, image processing and pick-and-place arms are some of the skills that we are evolving in. The level of expertise is increasing from just line-following and obstacle-avoiding to swarm robotic behav-

iour, which requires many robots to cooperate with each other to do a particular task.

Trends world over

Global trends also hint that preassembled robots are gaining more popularity and people usually use them as the first building block and develop applications over them, thus relieving them from the burden of building the chassis, finding a drive system and calibrating the system.

The scenario outside India is different mainly because people are ready to go for advanced robots and the technology and sensors used in them.

“Some kits like Bioloid enable the user to configure his robot in 18

different ways.—Jayakrishnan T., Director and Country Manager, Energid Technologies



They are ready to pay a high price for the same. Widely used robots are robotic arms, bipeds, humanoids and soccer robots.

Competitions are and have always been an incentive to make robots, be it in India or Japan. The difference lies in the level of the competition. *Robo Soccer* is a popular competition in which a team of four robots play a game of soccer against another team. Flying robot competition and humanoid robot fighting championship are some other popular events.

Advances in hardware and software

The electronic hardware keeps on advancing, nowadays in the form of complex sensors, communication systems and control systems. These technologies are definitely used in India but on a very small scale compared to other countries. However, a technology that is available in USA today can be made available in India tomorrow, thanks to global shopping sites like sparkfun.com and digikey.com.

Application-wise, we are overcoming the not-so-huge gap between us and other advanced nations.

A lot is happening in software development of the robotics. With the software available with robotic kits, the end user need not study the basics of microcontroller programming. He only needs to have a certain logical skill like what to do and what should be the end result.

“Basic functionalities like sensor interfacing, motor driving, servo driving and wireless command are in demand.—Pulkit Gaur, Founder and Chief Technology Officer, Gridbots Technologies

